

- x. If an Address Resolution Protocol (ARP) request is broadcast, an ARP reply is [1]
 a)Universal b)Unicast c)Multicast d)Generated locally
- Q.2 A** Explain the functions of different layers of OSI Model with suitable diagram [8]
- Q.2 B** Give the classification of transmission media & explain transmission impairments in detail. [4]
- Q.3 A** Explain working of selective repeat ARQ protocol with window diagram. [6]
- Q.3 B** What is bit and byte stuffing explain with example [4]
- Q.3 C** Differentiate between bridge and router.(Any 2 points) [2]
- Q.4 A** Explain ARP packet with suitable diagram. [4]
- Q.4 B** Draw a neat diagram of IPv4 header.Explain each field. [4]
- Q.4 C** From the given IP address **192.168.10.1/25**. Find the following [4]
 i) No. of networks
 ii) No. of IP addresses of each network
 iii) No. of host in each network
- Q.5 A** Draw UDP Header.The following is the dump of UDP header in Hexadecimal format <https://www.batuonline.com> [6]
CB 84 00 0D 00 1C 00 1C
 i) What is the source port number?
 ii) What is destination port number?
 iii) What is the total length of user datagram?
 iv) What is the length of data?
 v) Is the packet directed from a client to server or vice-versa?why?
- Q.5 B** Draw & explain connection establishment using 3 way handshaking in TCP. [4]
- Q.5 C** What is congestion? List TCP congestion control approaches. [2]
- Q.6 A** Write a Short Note on Electronic Mail (E-Mail) [4]
- Q.6 B** What is FTP? What are three file transmission modes? Explain them. [4]
- Q.6 C** What is DNS? Explain its working with suitable example. [4]
- Q.7** Write a Short Note on (4 Marks for each) [12]
 a) Digital Signature
 b) Entity Authentication
 c) Message Integrity

-----All the Best-----